



M003 Week 5 — English Worksheet

Topic: Layer Up — Terrain & Trees · **Course:** 3D Coordinates · **Time:** about 45 minutes

You already know the floor — (**x**, **z**), across and forward. This week you go **up**. One **fill** makes a solid box between **two corners**. Raise the **middle number (y)** and the box gets **taller**; stack wider and narrower boxes and flat ground turns into **trees, hills, and towers**.

Your mission this week: build this tree. A fat brown **trunk** going up, then a green **cap** of boxes that get **smaller** as they rise.

Build this



The idea

- A **3 × 3** brown **trunk**, stacked up the **y** axis.
- A green **cap** of 3 boxes: **widest** at the bottom, each one above is **narrower and higher**.
- Every box is one **fill** between two corners.

Read every coordinate the same way: (**x**, **y**, **z**)

1st number = X

across → (left ↔ right)

2nd number = Y

up ↑ (height)

3rd number = Z

forward ↗ (depth)

pos(**x** , **y** , **z**) → (**across** , **up** , **forward**)

Big idea: the **2nd number (y)** is height. Raise it and the box grows **up**. Stretch the **1st (x)** and **3rd (z)** to make a box **wider**.

1 · Predict

Read each set of steps. Look at each **column** — remember **1st = x**, **2nd = y**, **3rd = z**. Then write what you think you will see.

```
fill BROWN from (3, 0, 3) to (5, 6, 5)
```

X goes 3 → 5 and Z goes 3 → 5, so how wide is the trunk across and deep? Y goes 0 → 6, so how tall is it?


```
fill GREEN from (1, 7, 1) to (7, 8, 7)  
fill GREEN from (2, 9, 2) to (6, 11, 6)
```

Both are green boxes. Look at the x and z columns — which box is wider? Look at the y column — which box sits higher?


```
fill GREEN from (3, 12, 3) to (5, 14, 5)
```

This is the smallest cap box. Its y starts at 12 — is it near the top of the tree or the bottom?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — This should stack a **second cap box on top** of the first. Right now both boxes use the **same y**, so the second lands on the **same level** and nothing stacks up.

```
fill GREEN from (1, 7, 1) to (7, 8, 7)
fill GREEN from (2, 7, 2) to (6, 8, 6)
```

Hint: the **2nd number (y)** is height. Both boxes stop at $y = 8$. Which number must go higher?

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — This should build the **trunk going up** at $x = 3-5$, $z = 3-5$. Right now it makes a **flat slab** on the ground with no height.

```
fill BROWN from (3, 0, 3) to (5, 0, 5)
```

Hint: check the **2nd column (y)** in both corners.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — This green box should be a **wide cap** that sticks out past the trunk. Right now it is the **same width** as the trunk, so there is no cap.

```
fill GREEN from (3, 7, 3) to (5, 8, 5)
```

Hint: the trunk's x and z go 3 → 5. To overhang, the cap's **1st (x)** and **3rd (z)** must be wider.

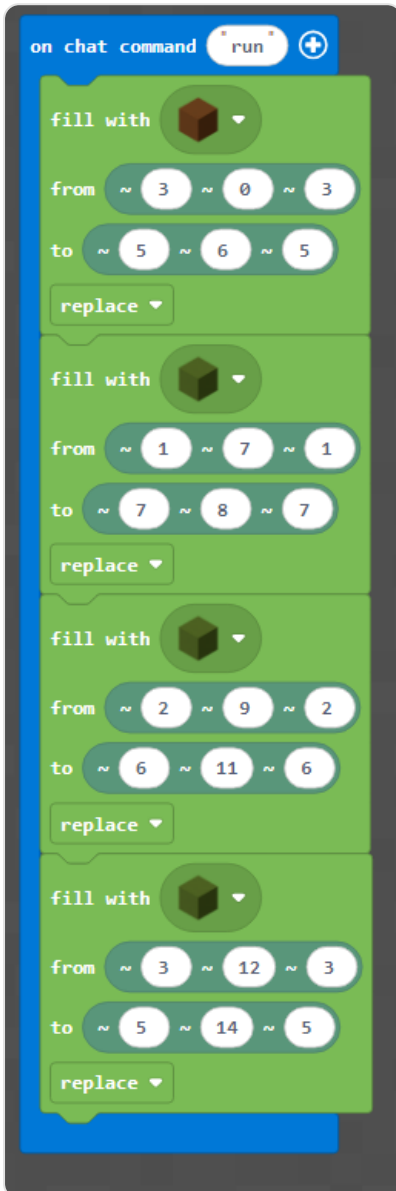
Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Build the Tree 📷

Now switch to your homework world and **build the tree from the top of this sheet**. Here is the exact recipe. Notice the colours: **x** · **y** · **z** in every corner.

Blocks



Python

```
def on_run():
    blocks.fill(BROWN_CONCRETE,
                pos(3, 0, 3), pos(5, 6, 5),
                FillOperation.REPLACE)
    blocks.fill(GREEN_CONCRETE,
                pos(1, 7, 1), pos(7, 8, 7),
                FillOperation.REPLACE)
    # ...two more cap boxes: y = 9-11, then y = 12-14
    player.on_chat("run", on_run)
```

When you finish, send a **photo or video** of your tree, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I built the trunk by ...

My trunk is ... tall and ... wide because the **x**, **y**, **z** numbers were ...

The number that grew for each new cap box was **y**, which means ...

My cap gets narrower because each box's **x** and **z** ...

One tricky moment was when ...

If I had more time, I would ...

First, I built the trunk by ...

My trunk is ... tall and ... wide because the **x**, **y**, **z** numbers were ...

The number that grew for each new cap box was **y**, which means ...

My cap gets narrower because each box's **x** and **z** ...

One tricky moment was when ...

If I had more time, I would ...

First, I built the trunk by ...

My trunk is ... tall and ... wide because the **x**, **y**, **z** numbers were ...

The number that grew for each new cap box was **y**, which means ...

My cap gets narrower because each box's **x** and **z** ...

One tricky moment was when ...

If I had more time, I would ...

First, I built the trunk by ...

My trunk is ... tall and ... wide because the **x**, **y**, **z** numbers were ...

The number that grew for each new cap box was **y**, which means ...

My cap gets narrower because each box's **x** and **z** ...

One tricky moment was when ...

If I had more time, I would ...

First, I built the trunk by ...

My trunk is ... tall and ... wide because the **x**, **y**, **z** numbers were ...

The number that grew for each new cap box was **y**, which means ...

My cap gets narrower because each box's **x** and **z** ...

One tricky moment was when ...

If I had more time, I would ...

First, I built the trunk by ...

My trunk is ... tall and ... wide because the **x**, **y**, **z** numbers were ...

The number that grew for each new cap box was **y**, which means ...

My cap gets narrower because each box's **x** and **z** ...

One tricky moment was when ...

If I had more time, I would ...

4 · Record Your Walkthrough

Take a video on your phone (or a parent's phone) while you show your tree in the world. Talk through it like you are teaching someone who has only seen flat pictures before. Try to use these words: **layer**, **fill**, **x**, **y**, **z**, **trunk**, **cap**.

1. Show your tree from the bottom of the trunk up to the top of the cap.
2. Point at one cap box and read its corner out loud — say which number is **x**, which is **y**, which is **z**.
3. Show two boxes and say which one is **wider** and why that makes a cap.
4. Say in your own words what each of **x**, **y**, and **z** does.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.